

SHIVANSHU SRIVASTAVA

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Professional Summary

Motivated and results-oriented B.Tech Computer Science student specializing in Python development, automated scripting, and backend engineering. Hands-on experience designing RESTful APIs with FastAPI and Flask, implementing RAG pipelines with LangChain and vector databases, and engineering robust web scrapers. Skilled in relational and non-relational database integration, Git-based workflows, and writing clean, object-oriented code. Quick learner with a problem-solving mindset seeking a Python Developer / Backend Engineer role.

Education

Sandip University

B.Tech, Computer Science and Engineering

2022 – Present

Nashik, Maharashtra

GlenHill School

SSC & HSC

2018 – 2021

Varanasi, Uttar Pradesh

Technical Skills

- Languages: Python 3.x, SQL, HTML/CSS, Bash
- Frameworks & Libs: FastAPI, Flask, Django, Pandas, NumPy, BeautifulSoup, Requests, LangChain
- Databases: PostgreSQL, SQLite, MongoDB, MySQL, SQLAlchemy ORM
- Tools & Platforms: Git, GitHub, VS Code, Postman, Linux/Ubuntu, Docker, Jupyter Notebook
- Core Concepts: REST APIs, OOP (Object-Oriented Programming), Data Structures & Algorithms, Web Scraping, Automation
- Testing: Pytest, Unit Testing, Debugging, Code Review

Internships

Brain O Vision

Machine Learning Intern

Jan 2024 – Mar 2024

Hyderabad, India

- Fine-tuned XGBoost and Random Forest models on consumer datasets to assist in improving predictive accuracy.
- Developed feature engineering pipelines and behavioral clustering models to preprocess customer data.
- Collaborated on a customer segmentation project using K-Means to identify distinct user cohorts for marketing analysis.

Code Alpha

Python Programming Intern

Nov 2025 – Nov 2025

Lucknow, India

- Wrote asynchronous Python scripts to automate routine tasks and optimize data processing workflows.
- Implemented unit tests to increase codebase reliability and identify bugs during active development cycles.
- Refactored older scripts into modular, object-oriented structures to improve code readability and maintenance.

Projects

TradeFlow – Trading Workspace

Live | Python, FastAPI, MongoDB, LangChain, Next.js

- Built a high-performance REST API with FastAPI supporting real-time data streaming and trade ingestion.
- Designed a flexible MongoDB schema with isolated per-user data architectures; integrated secure JWT authentication.

- Engineered an AI coaching module with LangChain and LLMs to parse execution logs and discover trading behavioral biases.

NSE Swing Screener

Live | Python, FastAPI, SQLite, BeautifulSoup, APScheduler

- Developed a quantitative stock screener scanning 500+ NSE stocks in parallel, lowering scanning runtime to under 60 seconds.
- Implemented Minervini Stage 2 and Volatility Contraction Pattern (VCP) detection mathematical algorithms on historical data.
- Utilized BeautifulSoup to scrape stock metrics and integrated SQLite as a local caching layer to optimize API rate-limits.

Autonomous Organizational Memory (ThirdEye)

Source | Python, ChromaDB, RAG, LangChain, FastAPI

- Created a Retrieval-Augmented Generation (RAG) platform to query system logs and corporate documentation.
- Integrated ChromaDB vector database for semantic search, custom text chunking, and similarity-based retrieval.
- Built FastAPI endpoints delivering natural language answers backed by custom source citations.

NSE Momentum Burst Scanner

Source | Python, BeautifulSoup, GitHub Actions, Telegram API

- Engineered an automated script running daily on GitHub Actions to scan Nifty 500 stocks for high-probability breakouts.
- Parsed HTML pages using BeautifulSoup to track stock trend confluences (RSI, ADX, MACD, and stochastic momentum).
- Integrated the Telegram Bot API to dispatch instant alert watchlists directly to active channel subscribers.

Hackathons

Regulatory Guideline Monitor & Updater (Top 10)

April 2026

Sunhacks 2026

- Developed a real-time, webhook-based compliance monitor tracking regulatory updates from NSE, BSE, MCX, and RBI.
- Engineered an interactive dashboard for visualizing policy updates and automating alerts for guideline changes.

HVAC Smart Control System (Top 10)

April 2024

Sunhacks 2024 by Sandip University & ESDS

- Designed a predictive model for HVAC energy optimization and climate control.

Sales Prediction & Product Performance Predictor Model

Oct 2023

AICTE Data Science Hackathon (Winner)

- Built regression models and performed feature engineering on 1000+ records.